

Detailed nerve endings

Scientists have used a technique called SNAP-tagging to produce the most detailed nerve images ever <http://dgit.in/SensoryNer>



Laser clean up

Tests are underway for cleaning rails on high-speed rail tracks using lasers by Dutch Railways <http://dgit.in/TrainLaser>

NERD WAR AMMUNITION

Of course, it is always a pleasure to know some bits of trivia about the spaceship. The next time your friends sit down for an intellectual discussion about the future of humanity, everyone should see you for the genius that you, dear reader, most certainly are.

- The total budget for the program will be north of \$350,000,000 per year of operation, which is about five times what our beloved Mangalyaan cost to put up in orbit near Mars. But considering that this one will carry and support humanity, we suppose we can understand the cost here. Although for reference, a US B2 bomber costs more than twice that amount. Food for thought.
- Your friendly neighbourhood Intel Core i7 processor is about 250 times faster than the processors used on board for this aircraft. Let that sink in – your beloved desktop is probably more computationally capable than something that is supposed to ferry humans 55.6 million kilometres and back. As interesting as that stat is, it always helps to remember the reason that it is so – all the heavy computation needed for the journey is the responsibility of Earth based computers, and the one on-board is supposed to mostly take

care of the navigation, and therefore really does not need all to sport all sorts of bells and whistles.

- The most important task of on-board computer(s) is verification of all the info that it is fed – nothing can ever cause an interruption. One of the most important parts of the avionics system for Orion is redundancy. There are twin processors in a single computer called the Flight Control Module that run in parallel, constantly checking against each other for differences. If anything is not 100% consistent actions are taken to immediately self-correct. Additional redundancy is added by having three identical Flight Control Modules (each with twin processors) running in parallel. This allows control to be switched to any of the backup FCMs if something causes one to require a reset. In the highly unlikely scenario that something renders all the FCMs ineffective there is a 5th completely different computer capable of running different code with a focus on crew survival and return functions.
- The systems make use of programming languages across almost all domains and specializations C, C++, C#, Python, Java, JavaScript, among many others. We explain these in a lot more detail for the interested in this month's FastTrack.

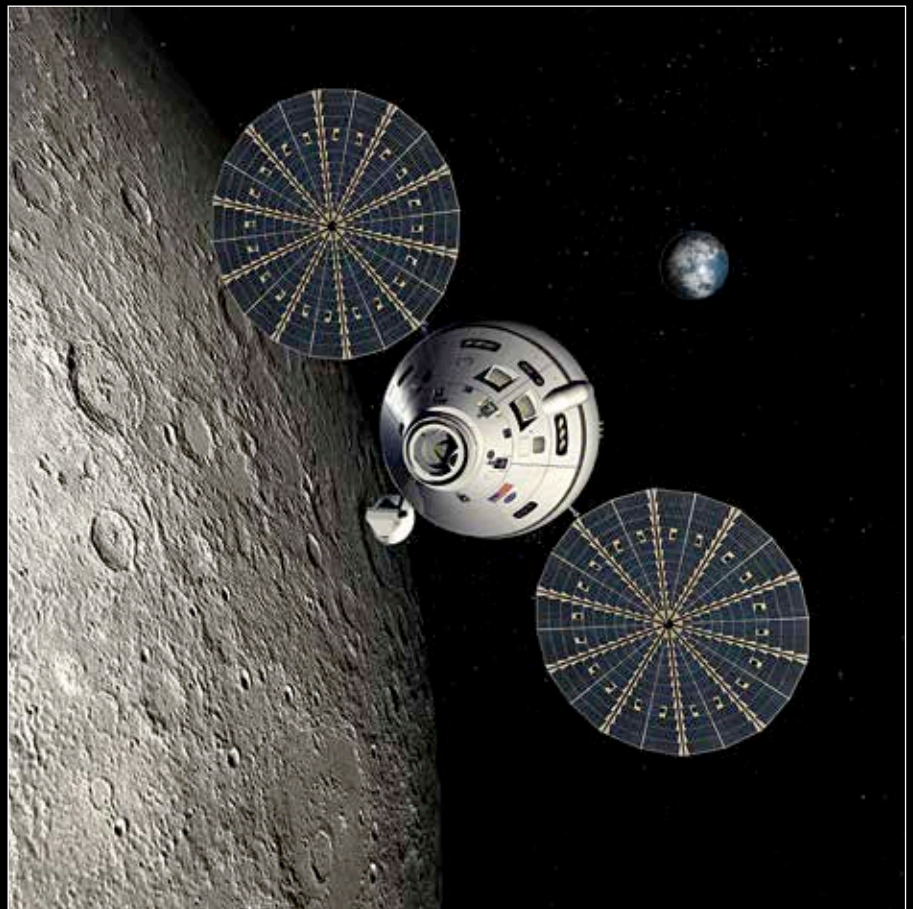
turn to Earth (yes, pretty much like what went down in *Interstellar*). It is planned to be followed by Exploration Mission 2.

Exploration Mission 2:

The continuation of the Exploration Mission series promises to be the most exciting part of the whole process. Exploration Mission 2 or EM-2 is scheduled to be the first manned mission of NASA's Orion project. This forms the core of the project that is supposed to restart manned exploration of the Solar System. With the developments and understanding of almost eight years of operation of the Orion modules, the assumption is that we should be capable of making humans reach so far beyond the Earth a reality. NASA has plans to launch it as early as 2021, taking a crew to a captured asteroid in lunar orbit. In the event this mission proceeds, it will be the first time humans have left low Earth orbit since Apollo 17 in December 1972.

Space Travel and Humanity

Space travel has had opinions divided among the general population like no other agenda. Some believe that as sentient life forms, it is imperative that we go on and explore our surroundings. Others



An artist's rendering shows the Orion crew capsule about to slingshot itself using the moon